

PulsePatch: Market, Solution, and Development Synopsis

Executive summary

PulsePatch is a smart wound patch that detects early infection and treats it automatically, aimed at people whose wounds are most likely to become infected.

Most everyday cuts heal fine: large emergency department studies put laceration infection rates at only 2.6% to 3.5%. The real problem is high-risk wounds, such as diabetic foot ulcers, surgical incisions, and chronic wounds in older adults. In these groups infections are common, costly, and can lead to hospitalization or amputation.

PulsePatch continuously measures wound pH, moisture, and temperature. A machine learning model reads the trends and flags rising infection risk. When risk rises, the patch runs low-frequency ultrasound to break up bacterial biofilms, then 405 nm violet light to kill the exposed bacteria. A phone app alerts the patient and, optionally, their care team.

The target market is home monitoring of high-risk wounds, the fastest-growing setting in a smart dressing market estimated at roughly \$0.9 to 2 billion in 2025 and growing about 6.5% to 11.5% per year.

The problem: high-risk wounds

High-risk wounds get infected far more often than ordinary cuts, and those infections drive hospital stays, amputations, and deaths. In the U.S. Medicare population alone, about 14.5% of beneficiaries (8.2 million people) had at least one wound or wound-related infection in 2014, at a conservative annual cost of about \$28 to \$32 billion ([Nussbaum et al., Value in Health 2018](#)). Surgical wounds and diabetic foot ulcers were the most expensive categories.

Group	How common	Infection risk	Consequences and cost
Diabetic foot ulcers	19% to 34% of people with diabetes develop one in their lifetime; about 33 million people affected worldwide	About 60% of ulcers become infected	15% to 20% of foot infections need amputation; advanced ulcers cost over \$50,000 per episode

Surgical site infections	About 300,000 per year in the U.S. (CDC toolkit via Becker's)	Infection within 30 days of surgery	Adds about 9.7 hospital days and over \$20,000 per admission
Chronic wounds in older adults	8.2 million Medicare beneficiaries	Pressure, venous, and arterial ulcers heal slowly and are prone to infection	Mean Medicare spending of about \$3,400 to \$3,900 per wound
Everyday lacerations (comparison)	Very common	2.6% to 3.5%	Mostly minor; diabetes is the biggest risk factor

Why early detection matters. Delayed presentation predicts poor outcomes for diabetic foot ulcers, and infections often go unnoticed because neuropathy removes pain as a warning sign. The same Diabetes Care review notes that temperature-sensing mats and socks detected problems weeks earlier than clinical exams, yet these at-home tools are not widely adopted. That gap is exactly where PulsePatch fits.

Market size and growth

Smart wound dressings are a roughly \$0.9 to 2 billion niche inside a \$25 billion global wound care market, and they are growing about twice as fast as the market overall. Estimates differ by research firm, so the range below is more reliable than any single figure.

Market	2025 size	Projected	Growth per year	Source
Global wound care	\$25.1B	\$35.1B by 2033	4.3%	Grand View Research
Global wound dressings	\$11.6B	\$15.9B by 2030	6.5%	MarketsandMarkets
U.S. wound care	\$7.5B	\$12.5B by 2035	5.4%	GMI

Smart wound dressings	\$2.06B	\$3.86B by 2035	6.5%	Metatech Insights
Smart wound care consumables	\$1.96B	\$7.27B by 2033	17.8%	Data Bridge
Smart bandages	\$0.88B	\$2.52B by 2034	11.5%	Towards Healthcare

Where the growth is. Within smart dressings, sensor-embedded dressings are the largest segment and home healthcare is the fastest-growing setting ([Metatech Insights](#)). Medicare wound spending is also concentrated in outpatient care rather than hospitals, which favors devices patients use at home.

Who pays. PulsePatch has three likely buyers:

- **Wound care clinics and podiatrists**, who manage diabetic foot ulcers and want to see between-visit changes.
- **Hospitals and surgical centers**, who are penalized for surgical site infections and readmissions and could send patients home with a patch.
- **Payers and home health agencies**, since one avoided amputation or readmission costs far more than a device.

Open question: whether PulsePatch could be billed under U.S. remote patient monitoring codes, which would make adoption much easier for clinics. This needs confirmation with a reimbursement specialist.

Competitive landscape

Existing products either detect infection during a clinic visit or treat wounds in the clinic; none we found both monitors continuously at home and treats automatically. That gap is PulsePatch's position.

Product	What it does	Where it's used	Gap PulsePatch fills
MolecuLight i:X and DX	Handheld fluorescence imaging that shows elevated bacterial burden; FDA-cleared Class II	Clinic, snapshot during visits	Continuous monitoring between visits, plus treatment

Swift Medical Swift Ray 1	Smartphone attachment combining photos, thermal imaging, and violet-light bacterial fluorescence	Clinic and home health nurses	Works without a clinician present; adds treatment
Sanuwave UltraMIST	40 kHz non-contact ultrasound through saline mist to clean wounds and reduce bacteria and biofilm; FDA-cleared	Clinic, mains-powered system	Wearable, low-power version triggered by sensor data
Research smart bandages (example)	pH and other sensors in a dressing, some with drug release	Mostly lab prototypes	Combines sensing with drug-free light and ultrasound therapy

Two takeaways matter for the pitch. First, established companies already use violet light and 40 kHz ultrasound for wounds, which validates PulsePatch's core technologies. Second, some insurers still classify non-contact low-frequency ultrasound as experimental ([BCBS policy](#)), so the monitoring and early-warning value should lead the pitch, with treatment as the differentiator.

Our solution: PulsePatch

PulsePatch is a wearable patch with reusable electronics and a disposable gel pad that senses infection risk and responds with drug-free ultrasound and violet-light therapy.

How it works

1. **Sense.** Sensors in the pad read wound pH, moisture, and temperature every few minutes.
2. **Predict.** A machine learning model compares the trends against the patient's own baseline and scores infection risk as normal, warning, or infection.
3. **Break the biofilm.** At warning or infection level, a piezo ultrasound transducer runs short pulsed cycles to disrupt the bacterial biofilm.
4. **Kill exposed bacteria.** 405 nm violet LEDs then run timed doses that kill a broad range of wound bacteria, including MRSA, without needing to identify the species.
5. **Alert.** The phone app notifies the patient and, if enabled, their clinician, and recommends an in-person visit if risk stays high.

Components

Layer	Part	Role
Shell	Flexible silicone housing, rechargeable battery, Bluetooth microcontroller (e.g., nRF52)	Runs the model, controls therapy, talks to the app; snaps off for reuse
Therapy	405 nm LED array; ultrasonic PZT disc (about 40 kHz for biofilm, optionally 1.5 MHz for healing stimulation) with driver circuit	Delivers light and ultrasound
Pad (disposable)	Hydrogel	Couples ultrasound and light into the wound; keeps it moist
Sensors	Polyaniline pH electrode, digital temperature chip on the wound plus one on healthy skin, printed moisture electrodes	Feed the model

Why these signals. Healthy wounds are mildly acidic, while infected wounds turn alkaline. One study treated pH 4 to 6 as normal, 6 to 7.5 as warning, and 7.5 to 9 as infection, and used a neural network to classify wound status from those readings ([Nano-Micro Letters 2025](#)). Local temperature rise is a well-validated early warning sign for diabetic foot problems ([Diabetes Care](#)).

Built-in safety. Therapy runs in short pulses, the temperature sensor acts as a heat cutoff, LEDs face only the skin, and dose limits cap daily light and ultrasound exposure.

What it is not. PulsePatch is for surface infections in high-risk wounds. It does not treat deep infections, bone infections, or tetanus, and does not replace antibiotics or a clinician.

Development plan

We are building PulsePatch in four phases, starting with sensing and a rule-based alert before adding machine learning and therapy, so each phase produces a working demo.

1. **Sensor prototype.** Build a breadboard version with a pH electrode, two temperature chips, and moisture electrodes on an nRF52 board, streaming to a phone app over Bluetooth. Test on synthetic wound fluid at pH 4 to 9 and on pork skin. Gate: stable, calibrated readings across the full pH range.

2. **Rules, then machine learning.** Ship simple threshold rules first (for example, pH above 7.5 plus temperature 1°C above the healthy-skin baseline). Meanwhile, build a labeled dataset and train a model. Gate: the model beats the rules on false alarms in testing.
3. **Therapy module.** Add the 405 nm LED array and a 40 kHz piezo transducer with a driver circuit and hydrogel coupling. Measure skin-surface heating and light dose, and test bacterial reduction on agar plates with a university microbiology lab. Gate: measurable bacterial reduction with temperature rise kept within safe limits.
4. **Wearable integration.** Move everything onto a flexible board in a silicone shell with a snap-off electronics pod and a disposable pad. Run wear tests for comfort, adhesion, and battery life. Gate: 24+ hours of wear with reliable data.

Machine learning approach

- **Data.** No large public dataset of labeled wound pH, moisture, and temperature exists, so we combine three sources: synthetic time series generated from published thresholds, our own lab recordings, and raw data requested from smart-bandage research groups (several offer data on request).
- **Features.** Current readings plus trends: the change in pH and temperature over the last few hours, and the difference from the healthy-skin sensor.
- **Model.** Start with a random forest in scikit-learn on Google Colab; move to a small time-series neural network only if needed.
- **Validation.** Use `TimeSeriesSplit` so the model is always tested on later data than it trained on, and `GroupKFold` so one wound's data never appears in both training and testing. Report false alarm and missed infection rates, not just accuracy.
- **Deployment.** Shrink the model with Edge Impulse or TensorFlow Lite for Microcontrollers so it runs on the patch itself, with the app as a backup.

Regulatory path. PulsePatch would be a medical device in the U.S. Comparable products, including MolecuLight's imaging devices and ultrasound wound cleaners like UltraMIST, are FDA Class II, which suggests a 510(k) pathway using them as predicates. A monitoring-only version could reach users sooner, with therapy added in a later submission. This needs confirmation with a regulatory advisor.

Key risks

- **Data.** Synthetic data may not match real wounds; real clinical validation will require a partner hospital or wound clinic and ethics approval.
- **Safety.** Ultrasound and LEDs generate heat near fragile skin, so dose limits and the thermal cutoff must be proven, not assumed.
- **Sensor reliability.** pH sensors drift and wound fluid varies; calibration and the bad-data detector are critical.
- **Reimbursement.** Some insurers treat low-frequency ultrasound wound therapy as experimental, which could slow adoption of the therapy features.

Sources

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